

Non-linear

Min

Max

Median

Smoothing of spatial filters

Linear

Box Filter

Weighted

Gaussian

Box Filter $\rightarrow \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (Mask)

Q

1	9	9	0	6
4	3	5	2	7
5	4	6	7	9
4	3	1	0	4
0	1	1	2	1

3x3 Mask

Now, multiply all the respective position with box filter

$$\Rightarrow \frac{1}{9} [3 \times 1 + 5 \times 1 + 2 \times 1 + 4 \times 1 + \dots]$$

$$= \frac{1}{9} [31] = 3.44 \approx 3$$

Now, the pixel value changes at middle i.e. 3.44.

Weighted Filter $\rightarrow$ It gives more weight to pixels near the output location.

$\left[\frac{1}{16} \right]$

1	2	1
2	4	2
1	2	1

For the above Q.

Ans: $3 + 10 + 2 + 8 + 24 + 14 + 3 + 2 = 66$

$\Rightarrow \frac{66}{16} = 4.1 \rightarrow \text{Round off } 4$

1	9	9	0	6
4	3	5	2	7
5	4	4	7	9
4	3	1	0	4
0	1	1	2	1

Median Filter: Suppose Filter

3 5 2
4 6 7
3 1 0

→ Sort → asc or desc.

0 1 2 3 4 5 6 7

Replace centre with 3

↓

3 5 2
4 3 7
3 1 0

For max, min filter simply
switch centre with max
max. value & min. value

Q Given below an image. Apply Median Filtering
using 3x3 mask. (Pixel Replication)

4	4	3	2	1	1
4	4	3	2	1	1
3	3	1	2	4	4
5	5	1	6	2	2
2	2	3	5	6	6
2	2	3	5	6	6

Image

Replications

First
horizontally then
vertically

Scanning:

Traverse through mask
from left to right.

First Mask → 4 4 3 Sort

(First row
included)

4 4 3

→ 4 4 4 4 3 3 3 1

3 3 1

Median = 3

2nd mask → Median → 33rd mask → Median → 24th mask → Median → 2

4 4 3

* Second row start: 3 3 1
5 5 1

5th Mask $\rightarrow$ Median $\rightarrow$ 3

6th $\rightarrow$ Median $\rightarrow$ 3

7th $\rightarrow$ Median $\rightarrow$ 2

8th $\rightarrow$ Median $\rightarrow$ 2

* Third Row start: 3 3 1
5 5 1
2 2 3

9th Mask $\rightarrow$ Median $\rightarrow$ 3

10th $\rightarrow$ Median $\rightarrow$ 3

11th $\rightarrow$ Median $\rightarrow$ 3

12th $\rightarrow$ Median $\rightarrow$ 4

* Fourth Row start: 5 5 1
2 2 3
2 2 3

13th Mask $\rightarrow$ Median $\rightarrow$ 2

14th Mask $\rightarrow$ Median $\rightarrow$ 3

15th $\rightarrow$ Median $\rightarrow$ 5

16th $\rightarrow$ Median $\rightarrow$ 6

Mask will always be odd.

(3x3, 5x5, 7x7, ...)
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Final Image :

4	4	3	2	1	1
4	3	3	2	2	1
3	3	3	2	2	4
5	3	3	3	4	2
2	2	3	5	6	6
2	2	3	5	6	6

Edge Detection: An edge can be defined as a set of connected pixels that form a boundary between two disjoint regions.

Healthcare
Biometric Ident.

2x2 * Robert Mask → works on diagonal edges only.

* ~~Sobel~~ & Prewitt → works on horizontal & vertical edges.

Image Mask →

20	20	50	50	50
20	20	50	50	50
20	20	50	50	50

Apply edge det. at (1,1)

Apply Robert Mask.

↓

2x2 Robert Mask Apply

Robert Mask $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

% Robert Mask at (1,1) → $(20 \times -1) + (20 \times 0) + (50 \times 0) + (50 \times 1)$
= 30

Sobel:

-1	-2	-1
0	0	0
1	2	1

PREWITT:

-1	-1	-1
0	0	0
1	1	1

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CORNER DETECTION :

* Harris corner detection

It is used to identify key locations in images for various applications like obj. recognition, img stitching & motion tracking

Horizontal $\rightarrow \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \quad \frac{\partial}{\partial x}$

Vertical $\rightarrow \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \frac{\partial}{\partial y}$

Corners are points where there are max. changes in intensity in both horiz. & vertical dir. within a local neighbourhood

- Harris matrix
- eigen values

* corner is intersection of two edges.

Input Image :

0	0	1	2	8	
1	0	5	7	11	1st
1	4	9	12	16	2nd
2	8	11	14	16	3rd
6	5	10	15	21	

Harris

corner Detection applied: Horizontal

1st Row

$$(1)(-1) + 0(0) + 1(5) = 4$$

$$0(-1) + 0(5) + 1(7) = 7$$

$$5(-1) + 0(7) + 1(11) = 6$$

2nd Row

$$1(-1) + 0(4) + 1(9) = 8$$

$$4(-1) + 0(9) + 1(12) = 8$$

$$9(-1) + 0(12) + 1(16) = 7$$

Horizontal Matrix = I_x $\begin{bmatrix} 4 & 7 & 6 \\ 8 & 8 & 7 \\ 9 & 6 & 5 \end{bmatrix}$

Vertical Matrix = I_y $\begin{bmatrix} 4 & 8 & 10 \\ 8 & 6 & 7 \\ 1 & 1 & 3 \end{bmatrix}$ $\left\{ \begin{array}{l} 0(-1) + \\ 0(0) + \\ 4(1) + \end{array} \right\}$

I_x^2 Horizontal = $16 + 49 + 36 + 64 + 64 + 49 + 81 + 36 + 25$
Sq. = 420

I_y^2 Vertical = $16 + 64 + 100 + 64 + 36 + 49 + 1 + 1 + 9$
Matrix = 340

Horix Vert
corresponding
only

$$\begin{bmatrix} 4 & 7 & 6 \\ 8 & 8 & 7 \\ 9 & 6 & 5 \end{bmatrix} \begin{bmatrix} 4 & 8 & 10 \\ 8 & 6 & 7 \\ 1 & 1 & 3 \end{bmatrix}$$

$I_x * I_y$

$\begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = \begin{bmatrix} 420 & 323 \\ 323 & 340 \end{bmatrix}$

$C = \det(H) - k \text{trace}(H)^2$	Harris	420	323
$k = 0.04$	Corner	323	340
	Score		

- * If C is large then it is corner (change in intensity in multiple direction)
- * If C is -ve then it is edge (change in intensity in one direction).
- * If C is small then it is a flat region (change in intensity in all dir).

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Corner Response

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$$\begin{array}{c|c} 420 & 323 \\ \hline 323 & 340 \end{array}$$

$$R = \det(H) - k \text{trace}(H)^2$$

$$C = ((420 \times 340) - (323 \times 323)) - 0.04 \left(\frac{420 + 340}{\text{Trace}} \right)^2$$

$\Rightarrow$ $\therefore$ Large answer = Corner

q

$$I = \begin{array}{ccc} 100 & 102 & 104 \\ 102 & 106 & 108 \\ 104 & 108 & 110 \end{array}$$

Solving using Sobel

$$S_x = \begin{array}{ccc} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{array}$$

$$S_y = \begin{array}{ccc} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{array}$$

Step I

$$S_x * I = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 100 & 102 & 104 \\ 102 & 106 & 108 \\ 104 & 108 & 110 \end{bmatrix}$$

$$= -100 + 104 - 204 + 216 - 104 + 110$$

$$= 4 + 12 + 6 = 22$$

Step II

$$S_y * I = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 100 & 102 & 104 \\ 102 & 106 & 108 \\ 104 & 108 & 110 \end{bmatrix}$$

$$= -100 - 204 - 104 + 104 + 216 + 110$$

$$= 10 + 12 = 22$$

Step II $I_x = 22$, $I_y = 22$
 $I_x^2 = I_y^2 = 22 \times 22 = 484$
 $I_x^2 = 484$
 $I_y^2 = 484$

Step III Harris Matrix $\begin{vmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{vmatrix} = \begin{vmatrix} 484 & 484 \\ 484 & 484 \end{vmatrix}$

Step IV Since we computed gradient only at center pixel,
 we don't have more pixels to average

~~Step IV~~ compute weighted sum over a small window (usually gaussian filter)

If not at centre pixel $\rightarrow \sum I_x^2$, $\sum I_y^2$, $\sum I_x I_y$

Step V ~~correct~~ Response : $R = \det(H) - k \text{trace}(H)^2$
 $= ((484 \times 484) - (484 \times 484)) -$
 $0.04 (484 + 484)^2$
 $= 0 - (968)(0.04) = -37480.96$
 $\therefore \text{-ve answer} = \text{Edge}$

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Morphological Operations: Geometric op. on image.

(1) DILATION → Pixels add w.r.t boundary

Vector addition & substitution.

$A \oplus B$

Increase brightness

{ Only check origin }

{ Substitution }

check for
poss. in the
selected
matrix

0 1 1 1	→	0 1 1 1	→	0 1 1 1	→	0 1 1 1
1 0 0 0		1 1 0 0		1 1 1 0		1 1 1 0
0 0 0 0		0 0 0 0		0 0 0 0		0 0 0 0

Vector Add. → A →	0 1 0	B →	1
	1 0 0		0 0
	0 0 0		

$\Rightarrow A = \{(0,1) (1,0)\}$
 $B = \{(0,0) (0,1)\}$
 $\left\{ \begin{array}{l} (0,1) + (0,0) = (0,1) \\ (0,1) + (0,1) = (0,2) \\ (1,0) + (0,0) = (1,0) \\ (1,0) + (0,1) = (1,1) \end{array} \right\}$ Put 1 at these pos.

0 1 1
1 1 0
0 0 0

Commutative prop. → $A \oplus B = B \oplus A$

Associative prop. → $A \oplus (B \oplus C) = (A \oplus B) \oplus C$

* Opening → Erosion followed by dilation

Closing → Dilation followed by erosion

we have to match a part of image with structuring element

(2) EROSION : It removes pixel

$$\{A \ominus B\}$$

It decreases brightness

we compare the structuring element and if the pixel structuring ele. matches then we replace the value by origin of structuring element.

Image :
(*)

1	1	1	1
1	1	0	1
0	1	1	1
0	1	0	0

Structuring
ele. (B)

1	1
1	0

origin value.

2x2
If 2x2 then this is origin value

Result :

0	1	0	0
0	0	0	0
0	1	0	0
0	0	0	0

When matched, replace the origin value and then all other values become 0

NO
commutative $\rightarrow A \ominus B \neq B \ominus A$
prop.

NO Associative $\rightarrow A \ominus (B \ominus C) \neq (A \ominus B) \ominus C$
prop.

* Opening :

1	1	1	0	1	1	1
1	①	1	1	1	①	1
1	1	1	0	1	1	1

(A)

1	1	1
1	①	1
1	1	1

(B)

Erosion →

0	0	0	0	0	0	0
0	①	0	0	0	①	0
0	0	0	0	0	0	0

origin is ^{center} ~~center~~ in
2x3

Performing Dilation : } ∵ center i.e. origin matches
put B completely in parts
of A resp.

⇒

1	1	1	0	1	1	1
1	1	1	0	1	1	1
1	1	1	0	1	1	1

1 → white
0 → Black

* Closing :

1	1	1	1	1	1	1
1	①	①	0	①	①	1
1	1	1	0	1	1	1

(A)

1	1	1
---	---	---

1	①	1
---	---	---

1	1	1
---	---	---

(B)

Dilation : ① → circled origins are matching
thus put entire (B) in those resp.
parts of A.

Result →

1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1

} Substitution
step

Erosion -